

# Physics Simulation and STRidge Identification Report

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.

## 1. Simulation Parameters

| Quantity                      | Symbol     | Value           | Unit                           |
|-------------------------------|------------|-----------------|--------------------------------|
| Fluid density                 | $\rho$     | 9.200000000e+02 | kg m <sup>-3</sup>             |
| Dynamic viscosity             | $\nu$      | 8.100000000e-02 | Pa s                           |
| Kinematic viscosity           | $\nu$      | 8.80434783e-05  | m <sup>2</sup> s <sup>-1</sup> |
| Characteristic velocity       | $u_{\max}$ | 1.000000000e+00 | m s <sup>-1</sup>              |
| Characteristic length         | $D$        | 5.000000000e-03 | m                              |
| Reynolds number               | $Re$       | 1.97740055e+01  | dimensionless                  |
| Body force                    | $G$        | 9.810000000e+00 | m s <sup>-2</sup>              |
| Grid points                   | $N_x$      | 3.100000000e+01 | points                         |
| Unused transverse grid points | $N_y$      | 3.100000000e+01 | points                         |
| Grid spacing                  | $\Delta y$ | 1.61290323e-04  | m                              |
| Time step                     | $\Delta t$ | 1.31646880e-04  | s                              |
| Numerical threshold           | $\epsilon$ | 1.000000000e-15 | -                              |

## 2. Numerical Discretisation

| Quantity | Symbol | Value | Unit |
|----------|--------|-------|------|
|----------|--------|-------|------|

Parameters listed above are the values used by the selected solver.

## 3. Data and Pre-processing

| Quantity                        | Value      |
|---------------------------------|------------|
| Loaded field shape $(t, y)$     | (2142, 31) |
| Regression field shape $(t, y)$ | (542, 27)  |
| Regression samples              | 14,634     |
| Dictionary terms                | 7          |
| First saved time index          | 150        |
| Temporal crop at each boundary  | 800 layers |

| Quantity        | Value    |
|-----------------|----------|
| x-boundary crop | 0 points |
| y-boundary crop | 2 points |

## 4. Governing Equation Verification

Reference equation

$$u_t = 8.80434783e-05 * u_{yy} + 9.81000000e+00 * one$$

Learned equation

$$u_t = - 2.08702206e-10 * u * u_y + 2.68851484e-07 * u + 1.56320160e-08 * u_y + 8.84726794e-05 * u_{yy} + 9.85782285e+00 * one$$

| Term           | Reference coefficient | Learned coefficient | Absolute error   |
|----------------|-----------------------|---------------------|------------------|
| $u * u_y$      | $+0.00000000e+00$     | $-2.08702206e-10$   | $2.08702206e-10$ |
| $u * u_{yy}$   | $+0.00000000e+00$     | $+4.66507141e-12$   | $4.66507141e-12$ |
| $u_y * u_{yy}$ | $+0.00000000e+00$     | $+1.34869663e-13$   | $1.34869663e-13$ |
| $u$            | $+0.00000000e+00$     | $+2.68851484e-07$   | $2.68851484e-07$ |
| $u_y$          | $+0.00000000e+00$     | $+1.56320160e-08$   | $1.56320160e-08$ |
| $u_{yy}$       | $+8.80434783e-05$     | $+8.84726794e-05$   | $4.29201163e-07$ |
| 1              | $+9.81000000e+00$     | $+9.85782285e+00$   | $4.78228475e-02$ |

## 5. STRidge Selection and Error Metrics

| Metric                              | Value            |
|-------------------------------------|------------------|
| Regularisation parameter, $\lambda$ | $1.00000000e-09$ |
| Selected tolerance                  | $1.00000000e-06$ |
| L0 penalty                          | $1.00000000e-06$ |
| Training RMSE                       | $2.63621763e-09$ |
| Validation RMSE                     | $3.34331075e-09$ |

## 6. Reproducibility Notes

- Input directory:  
`/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_2movingwalls/u_velocity_field`
- Fields were loaded starting at saved time index 150.

- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.
- The learned equation is compared term-by-term with the supplied reference coefficients.